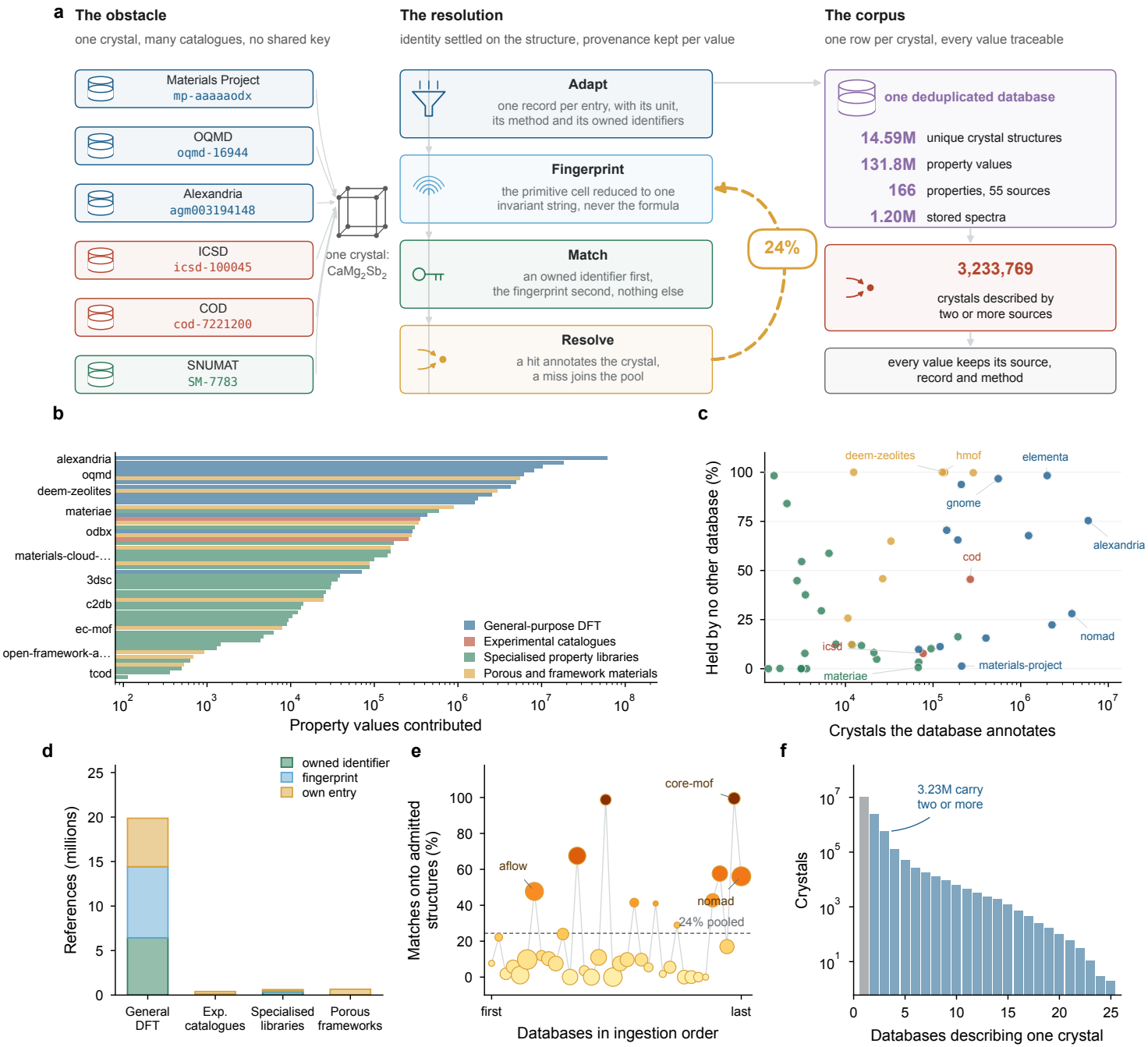

a
The obstacle
one crystal, many catalogues, no shared key
Materials Project
mp-aaaaaodx
OQMD
oqmd-16944
Alexandria
agm003194148
ICSD
icsd-100045
COD
cod-7221200
SNUMAT
SM-7783
one crystal:
CaMg2Sb2
The resolution
identity settled on the structure, provenance kept per value
Adapt
one record per entry, with its unit, its method and its owned identifiers
Fingerprint
the primitive cell reduced to one invariant string, never the formula
Match
an owned identifier first, the fingerprint second, nothing else
Resolve
a hit annotates the crystal, a miss joins the pool
24%
The corpus
one row per crystal, every value traceable
one deduplicated database
14.59M unique crystal structures
131.8M property values
166 properties, 55 sources
1.20M stored spectra
3,233,769
crystals described by two or more sources
every value keeps its source, record and method
b
alexandria
oqmd
deem-zeolites
materiae
odbx
materials-cloud-…
3dsc
c2db
ec-mof
open-framework-a…
tcod
General-purpose DFT
Experimental catalogues
Specialised property libraries
Porous and framework materials
Property values contributed
c
deem-zeolites
hmof
elementa
gnome
alexandria
cod
nomad
icsd
materiae
materials-project
Held by no other database (%)
Crystals the database annotates
d
owned identifier
fingerprint
own entry
References (millions)
General DFT
Exp. catalogues
Specialised libraries
Porous frameworks
e
core-mof
aflow
nomad
24% pooled
Matches onto admitted structures (%)
first
last
Databases in ingestion order
f
3.23M carry two or more
Crystals
Databases describing one crystal